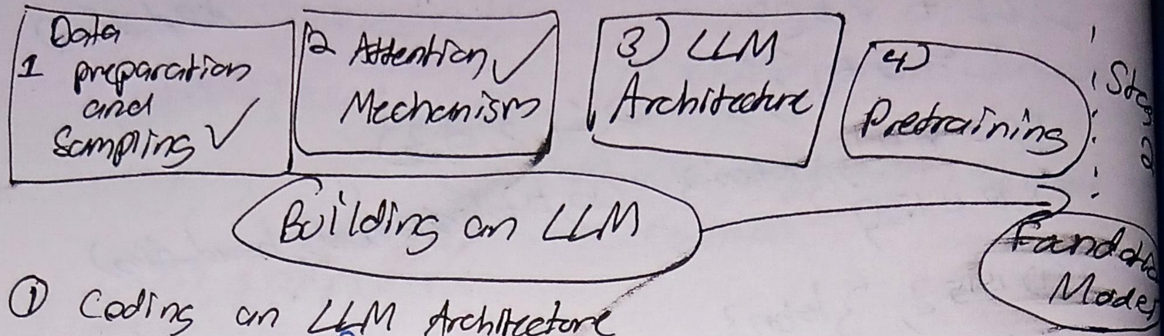
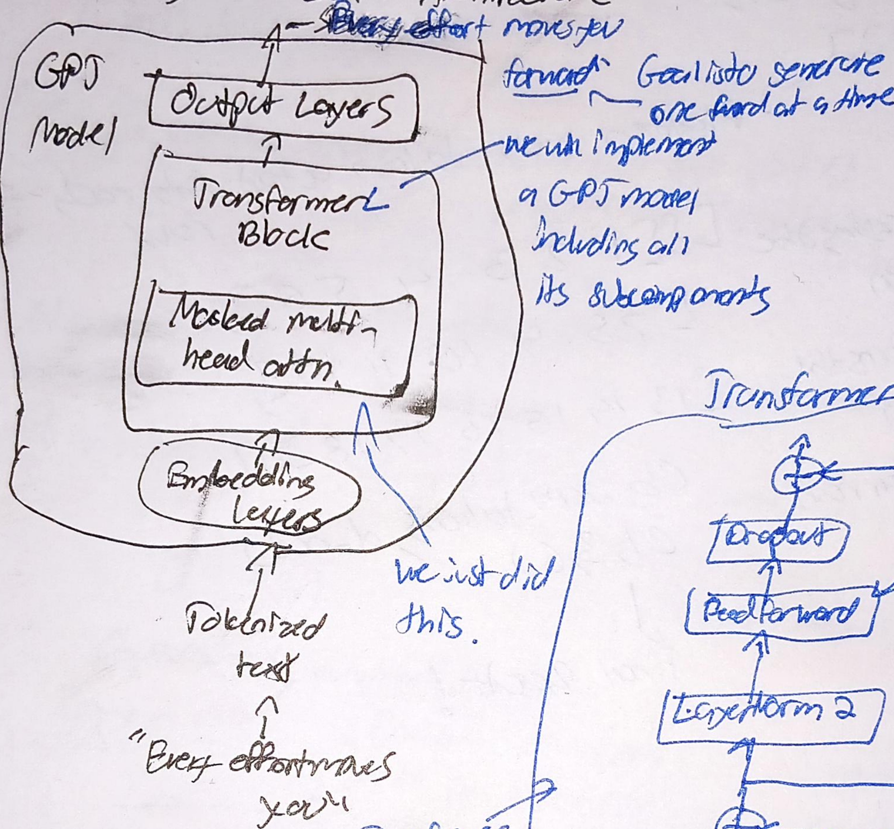
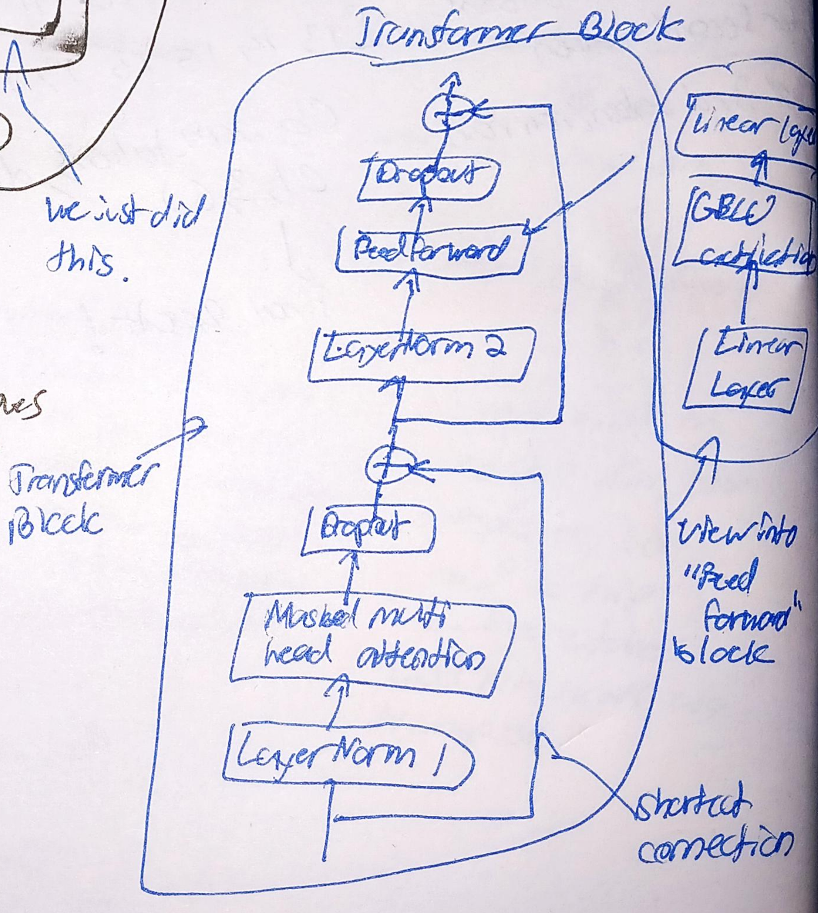
Birds eye view of LLM architecture

Stage 1



① Coding an LLM Architecture





② What have I learned?

(a) Input Tokenization

(b) Embedding (Token + Positional)

(c) Masked multi-head attention

③ Whats next?

(a) Transformer blocks

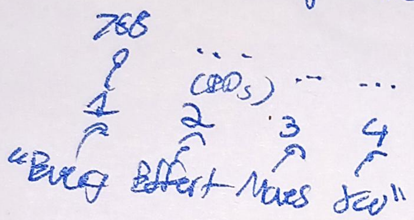
④ Scale upto size of a small GPT-2 model -> 124 million parameters

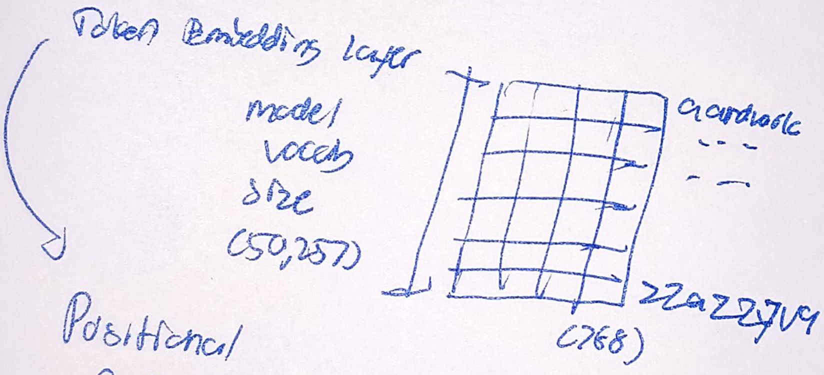
⑤ OpenAI has made GPT-2 weights public. GPT-3,4 weights have not yet been made public

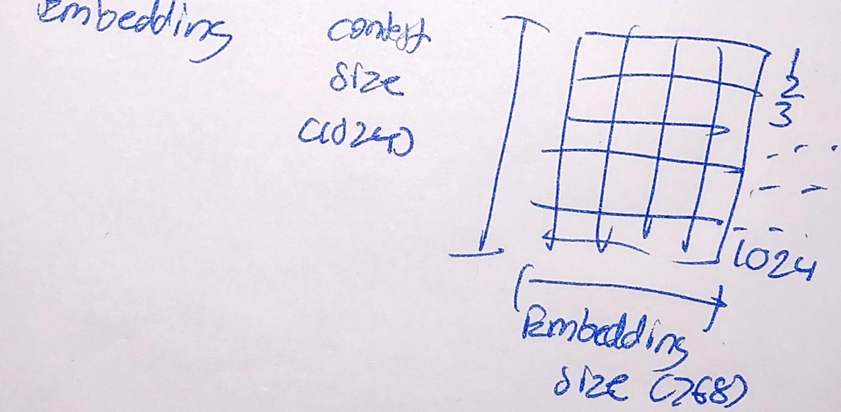
```
GPT_config_124M = {
  'vocab_size': 50257
  'context_length': 1024
  'emb_dim': 768
  'n_heads': 12
  'n_layers': 12
  'drop_rate': 0.1
  'qkv_bias': False
}
```

↳ Using this config, build a GPT placeholder arch. (dummy GPT)

↳ Give us Birds eye view of how everything fits together.







output logits for 1 test sample:

Token 1 ---- 40000 ⊙ each ele. corresponds to probability of being next word

Token 2 ----

Token 3 ----

Token 4 ----

vocab size 50257

In the end,
we use multiple
transformer blocks
to implement GPT
model that we will train in
upcoming chapter

7) Final GPT
Architecture

Next, we implement the
blocks 2-5 that
are in GPT
model

6) Transformer
block

combine building blocks 2-5
including the multi-head
attention module covered in
the previous chapter, into a
transformer block.

2) Layer
Normalization

3) GELU
Activation

4) Feed forward
network

5) Shortcut
connections

1) GPT backbone

Start this chapter
by implementing a
GPT placeholder model
to see the overall
structure of the model